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INTRODUCTION OF DATA ANALYTICS

Data Analytics Lifecycle • Types of Data • Real-World Examples

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Today’ Agenda

01 **What is Data Analytics?**
Definition, importance & real-world relevance

02 **Data Analytics Lifecycle**
6-phase process from problem to insight

03 **Types of Analytics**
Descriptive, Diagnostic, Predictive, Prescriptive

04 **Types of Data**
Structure, Availability, Measurement and Observation

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What is Data Analytics?

Data Analytics is the process of examining, cleaning, transforming, and modelling data to **discover useful information, draw conclusions, and support decision-making.**

🔍 Think of it like this:

A doctor looks at your test reports (data), analyses symptoms (patterns), and recommends treatment (decision). That's data analytics in real life!

📈 Why Does It Matter?

- Companies make smarter decisions
- Detect fraud and errors early
- Understand customer behaviour
- Predict future trends

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Data Analytics Lifecycle

The 6-Phase Process Every Data Analyst Follows

1. Discovery



Understand the business problem & define goals

2. Data Collection



Gather data from databases, APIs, surveys, files

3. Data Preparation



Clean, transform & format data for analysis

4. Data Analysis



Apply statistical methods & explore patterns

5. Visualisation



Create charts, dashboards & visual stories

6. Decision & Action



Share insights, drive business decisions

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Lifecycle in Action — Real Example

🛒 **Scenario: An e-commerce company wants to reduce product return rates.**

Discovery	Why are customers returning products? Is it size, quality, or wrong items?
Data Collection	Collect return records, customer feedback forms, product reviews & order history.
Data Preparation	Remove duplicate returns, fix missing reason codes, standardise product categories.
Data Analysis	Find: 62% returns are due to size mismatch. Most returns are from mobile shoppers.
Visualisation	Create a dashboard showing return % by category, device, and region.
Decision & Action	Add a size guide on mobile, launch a virtual try-on feature → returns drop 35%!

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Types of Analytics



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Types of Analytics - Real Example

From understanding the past → shaping the future



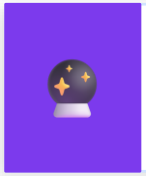
Descriptive Analytics — "What happened?"

Swiggy notices that delivery delays increased in a particular area this week.



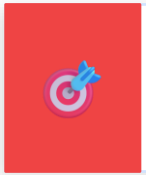
Diagnostic Analytics — "Why did it happen?"

Swiggy analyzes the reason and finds heavy traffic and rain caused the delays.



Predictive Analytics — "What will happen?"

Swiggy predicts that delays may continue during evening peak hours.



Prescriptive Analytics — "What should we do?"

Swiggy suggests alternative routes and assigns more delivery partners to reduce delays.

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Types of Data

Structure

Observation

Measurements

Availability

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Types of Data : Structure

Structured Data



Organised in rows & columns (like a spreadsheet or database table).

Examples:

- Excel sheets
- SQL databases
- Customer records
- Financial reports

Your bank statement — each transaction is a row with date, amount, description.

Semi-Structured



Has some structure but doesn't fit neatly in a table. Contains tags or markers.

Examples:

- JSON files
- XML data
- Emails
- HTML pages

A WhatsApp message export — has sender, time, text but not in a fixed table.

Unstructured Data



No predefined format or model. The majority of the world's data is unstructured.

Examples:

- Photos & videos
- Audio files
- Social media posts
- PDF documents

Customer reviews on Amazon — free-form text with no fixed columns.

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Types of Data : Observation

Cross-Sectional Data



Observations of many subjects at one point in time.

Examples:

- Heights of students in a class in 2026
- Salaries of employees in a month
- Marks of different students in a single exam

A school records marks of all students in one exam conducted today.

Time Series



Observations of one subject across different time periods.

Examples:

- Daily temperature of city over 1 month
- Stock price over one year
- Yearly GDP of India from 2010 to 2025

A company tracks its monthly profit from January to December

Panel Data



Observations of many subjects across multiple time periods.

Examples:

- Monthly sales of multiple stores over 12 months
- Yearly income of different countries over 10 years
- Performance scores of multiple students tracked across semesters

Monthly sales of 10 stores for 3 years.

Types of Data : Observation

Cross Sectional Data

Year	City	Average Sales (Crore INR)
2023	Chennai	500
2023	Delhi	600
2023	Mumbai	550

Time Series Data

Year	Average Sales in Chennai (Crore INR)
2020	480
2021	510
2022	520

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Types of Data : Observation

Panel Data

Year	City	Average Sales (Crore INR)
2020	Chennai	480
2020	Delhi	560
2020	Mumbai	520
2021	Chennai	510
2021	Delhi	590
2021	Mumbai	540
2022	Chennai	520
2022	Delhi	610
2022	Mumbai	560

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Types of Data : Measurement

📊 Quantitative Data (Numbers)

Data that can be measured and expressed as numbers.

Discrete (countable):

- Number of students: 35
- Number of orders: 142
- Number of clicks: 5,000

> 💡 *Number of students in a classroom, which can be counted.*

Continuous (measurable):

- Height: 5.9 feet
- Temperature: 36.6°C
- Sales revenue: ₹1,24,500

> 💡 *Temperature or height, which can take any value within a range.*

🗣️ Qualitative Data (Categories)

Data that describes characteristics or qualities — not numbers.

Nominal (no order):

- Gender: Male / Female
- City: Chennai, Mumbai, Delhi
- Product: Laptop, Phone, Tablet

> 💡 *Customer categories like Male, Female, and Other without any order.*

Ordinal (has order/rank):

- Rating: Poor < Average < Good
- Education: School < UG < PG
- Size: Small < Medium < Large

> 💡 *Customer satisfaction ratings like Poor, Good, Better, Excellent with a meaningful order.*

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Types of Data : Availability

Primary Data



Collected first-hand by the researcher for a specific purpose. Fresh, relevant, expensive.

Examples:

- Surveys
- Interviews
- Questionnaires
- Direct observations

💡 *A company collecting customer feedback through a Google Form*

Secondary Data



Already collected by someone else. Cheaper, quicker — but may not perfectly fit your needs.

Examples:

- Government reports
- Research papers
- Customer records
- Online articles and databases

💡 A startup using census population data for market analysis

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Data Sources

Open / Public Data

Freely accessible to anyone. Drives innovation.

Data from WHO, Kaggle datasets, data.gov.in — all free, used by millions of apps.

Private / Proprietary Data

Restricted to authorized users. Highly valuable, legally protected.

Bank's transaction logs.. Aadhaar biometric data — never shared publicly.

Commercial / Paid Data

Sold by vendors. High quality, updated regularly.

Bloomberg terminal data used by SEBI analysts. Subscription-based business analytics data

Key Takeaways

🎓 What We Covered Today

- Data Analytics turns raw data into useful insights
- The 6-phase lifecycle guides every analysis project
- 4 analytics types: Descriptive → Diagnostic → Predictive → Prescriptive
- Data types Break Down

Great start! Every expert was once a beginner. Keep learning, keep analysing! 💪